

Deep Learning for Master Students Mandatory coding task

1 Setup

- run `python3 -m venv venv` inside the project root folder in the terminal to create a virtual environment and start the virtual environment
- run `pip install -r requirements.txt` in the terminal
- unzip the images (e.g. into `your_project_root/coding_task_data/EuroSAT_RGB` and `your_project_root/coding_task_data/EuroSAT_MS`)
- go into `config.yml` and set the following parameters:
 - `project_root = "your_project_root" # "." should work`
 - `dataset_root = "your_dataset_root"` (if the datasets are saved as described, `"/coding_task_data/EuroSAT_RGB"` and `"... /EuroSAT_MS"` should work)

2 Structure

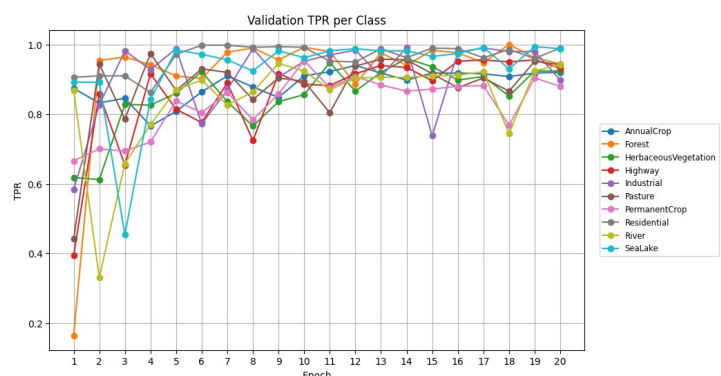
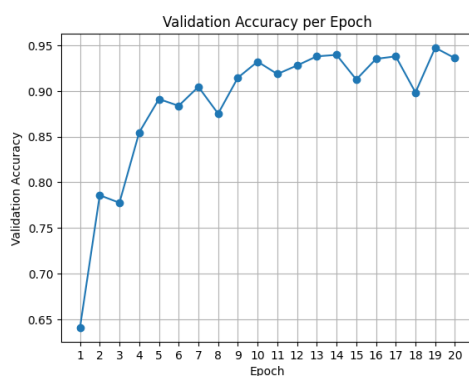
1. Coding Task 1 is solved inside `./data_splitter.py`
2. To run the prediction on the test data using the final model, run `./predict.py` (This saves the logits every time)
3. To run the reproduction routine using the final model, run `./reproduce.py` (To save the logits in the initial run, set `"save_logits: True"` inside the `config.yml`. To compare the logits, set `"save_logits: False"`)

3 Results

Each model was trained on 10.000 training images, 5000 validation images and tested on 5000 test images. The training was run with a batchsize of 16 in 625 batches for 20 epochs with a learning rate of 0.0001.

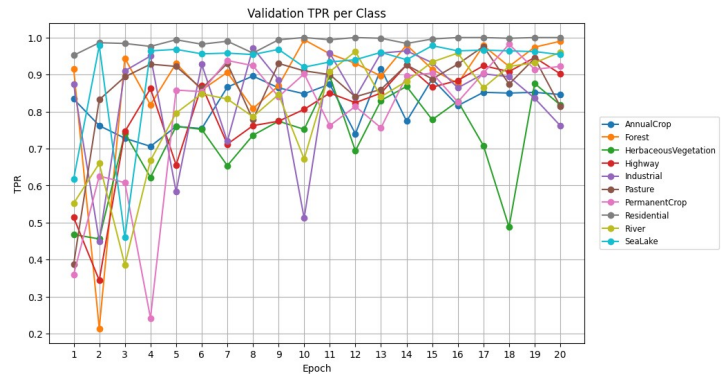
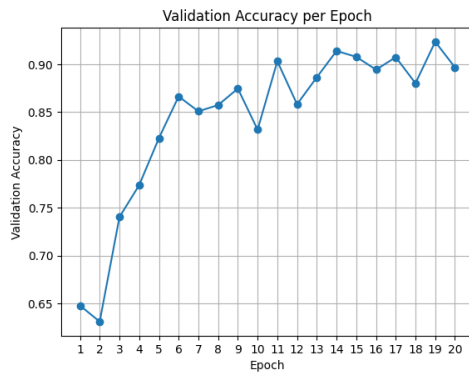
3.1 EuroSAT RGB

3.1.1 Mild Data Augmentation



Best model: Epoch 19 – Validation Accuracy: 0.9474

3.1.2 Strong Data Augmentation



Best model: Epoch 19 – Validation Accuracy: 0.9236

3.1.2.1 Finally selected model

Mild Data Augmentation

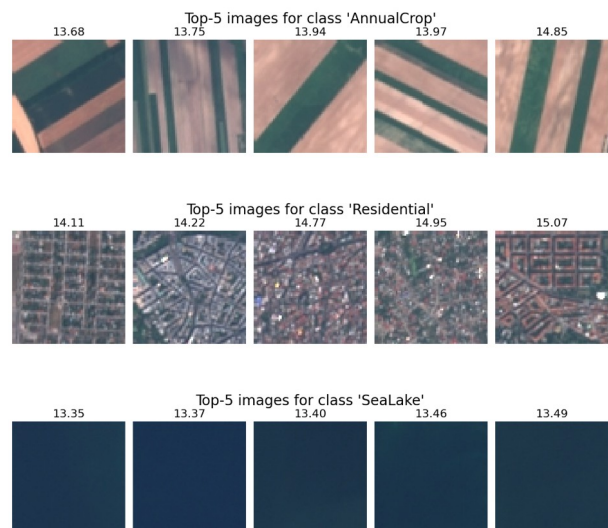
- Test Accuracy: 0.9336
- Test TPR: {0: 0.942, 1: 0.938, 2: 0.912, 3: 0.93, 4: 0.88, 5: 0.9, 6: 0.898, 7: 0.996, 8: 0.952, 9: 0.988}

3.1.2.2 Ranking

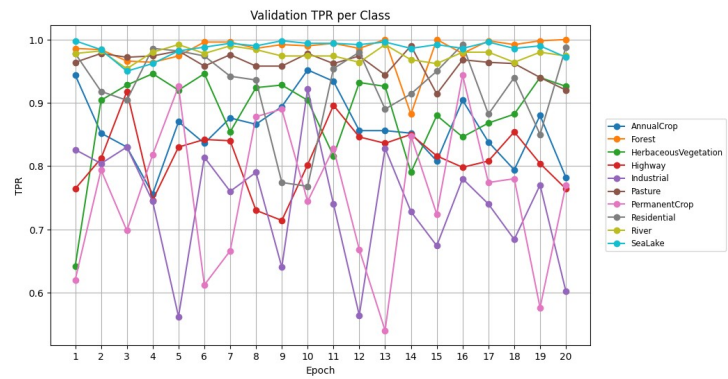
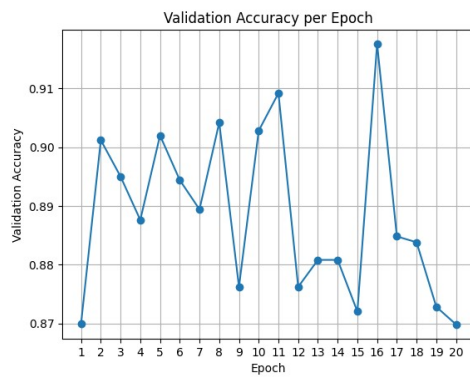
Bottom Images



Top Images



3.1.3 EuroSAT MS

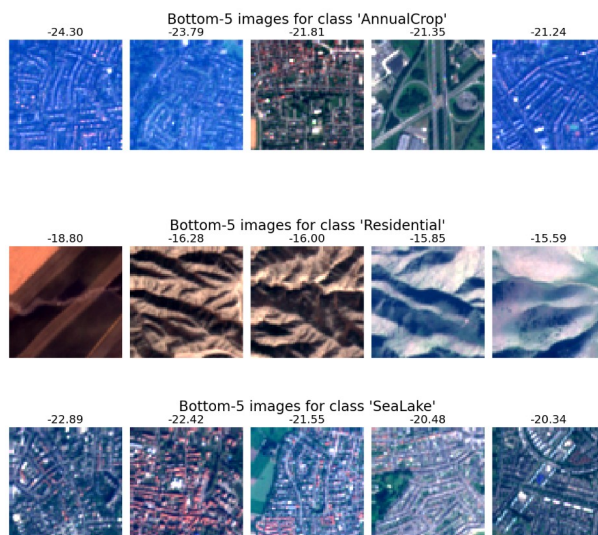


3.1.3.1 Finally selected model

- Epoch 16 – Validation Accuracy: 0.9176
- Test Accuracy: 0.8698
- Test TPR: {0: 0.784, 1: 1.0, 2: 0.92, 3: 0.784, 4: 0.58, 5: 0.91, 6: 0.768, 7: 0.988, 8: 0.982, 9: 0.982}

3.1.3.2 Ranking

Bottom Images



Top Images

